# Robed Beauvile-Owen, Ph.D.

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## EXPERIENCE

**Founder & Data Scientist** — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy for retail and service businesses and identified $1.5M+ expansion opportunities.
- Designed a revenue driver tree (traffic × capture rate × ticket price) and a margin model that turned client profit goals into break-even targets.
- Estimated the incremental revenue of new sites with observational causal inference (Double ML, causal forests) on a 12-year county panel, with robustness checks (difference-in-differences) and out-of-sample back-tests against real sites.
- Developed scenario planning (low/base/high) and Monte Carlo risk models (probability of loss, VaR, CVaR) on reproducible, tested Python workflows (150+ unit tests) that cut client report turnaround by 91%.

**Senior Data Scientist** — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Owned the consumer funnel KPI framework and driver tree linking gross lead rate (GLR) to member rate and cost per member with strategy, finance, and engineering, and validated GLR as the primary lever with a causal graph (DAG) tested on historical data.
- Translated financial goals into team targets with a funnel sensitivity model showing the cost-per-member target required GLR of at least 14.2%.
- Built a certified GLR definition for a new client whose data lacked key lead markers, working with data engineering to date leads by last response, credit them to campaigns, and flag broker influence.
- Developed uplift and segmentation models on millions of HIPAA-protected records, used in 87% of campaigns, cutting ad spend by $10K per campaign and moving 12% of one client's budget to its top GLR segment.
- Created an in-house experimentation and reporting platform that data science and strategy used to validate lift and to run monthly reporting, weekly analysis, Power BI ingestion, and metric monitoring.
- Found that claims and ticket complaints drove churn and added them to the churn model, then re-architected its platform on Spark and Databricks, cutting runtime by 48% and memory by 23%.

**Founding Machine Learning Engineer** — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, defined success metrics and drift monitoring for search and pricing models, and mentored the team on ML practices.
- Shipped a regression pricing API (C#, Azure, 10,000+ daily requests) that cut false and anomalous shipping prices by 13%, reducing cost leakage and holding a stable 18% contribution margin.
- Cut search dissatisfaction by 12% and raised platform engagement by 16% with custom Elasticsearch analyzers, field boosting, and relevance tuning.

**Data Scientist** — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Raised the XGBoost GLR model's AUC-ROC 3% relative to the prior model and validated it with an A/B test. SHAP surfaced the drivers (income, age, children in the household) and uplift models isolated the creative's effect.
- Validated GLR out of sample as a leading indicator of member rate 60 days ahead ($R^2 = 0.28$). Model-driven targeting and creative changes raised gross lead rates by 18%.
- Led monthly monitoring of up to 10 production models across clients (accuracy, drift, forecast vs. actual) for leadership, and built a company-wide self-service dashboard on curated Snowflake and Databricks tables.

## TECHNICAL PROJECTS

**Customer LTV & Churn Survival Modeling** github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Simulated 1M customer lifetimes with tenure-dependent churn (10%, 5%, then 2% monthly) to show the LTV = ARPU / churn identity holds only under constant churn.
- Estimated expected lifetime and LTV from the empirical survival curve ($E[T] = \int S(t)dt$), matching the simulated mean.

**Mobile Game Retention A/B Test** github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Analyzed a test moving a mobile game's first progress gate from level 30 to 40, measuring 1- and 7-day retention and rounds played.
- Set Type I/II error rates, sized samples, removed outliers with IQR, and weighed statistical against practical significance.

**Medicare Claims Intelligence & Fraud Detection** github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a dbt warehouse (4 staging models, 5 marts, 26 data tests) over 790K CMS claims, with tests that caught 11,253 claims missing a service date.
- Beat a seasonal-naive forecast in a 35-origin rolling backtest (MASE 0.62 vs. 1.06), reached 81% precision on planted fraud at review capacity, and built an out-of-time XGBoost risk model (SHAP, MLflow) finding 46% more high-cost members.

## EDUCATION

**Ph.D. in Mathematics** — *University of Pittsburgh* 2023

**Dissertation:** A Minimal Invariant Set, Frames, and Operators

**B.S. in Mathematics** — *Florida State University* 2014 - 2017

**Minor:** Philosophy | **GPA:** 3.90

## TECHNICAL SKILLS

**Causal Inference & Experimentation:** A/B Testing, Experimentation Platforms, CUPED, Power Analysis, Observational Causal Methods, Double ML, Causal Forests, Difference-in-Differences, Uplift Modeling, Causal Graphs (DAGs), Robustness Checks

**Modeling & Statistics:** Regression, Classification, XGBoost, Time Series Forecasting, Survival Analysis, Monte Carlo Simulation, Graph Models, Multi-Metric Modeling, SHAP, Model Monitoring & Drift

**Data & Tools:** SQL, Python (pandas, statsmodels, scikit-learn, econml), R (survey analysis), C#, Snowflake, Databricks, Spark, dbt, Curated Tables & Data Marts, AWS, Azure, Power BI, Git

**Collaboration:** Cross-Functional Partnership (Product, Finance, Engineering, Strategy), Stakeholder Communication, Method Documentation, Mentoring

**Metrics & Planning:** KPI Frameworks, KPI / Driver Trees, Certified Metric Definitions, Leading Indicators, Consumer Product Analytics (Acquisition, Activation, Engagement, Retention, Monetization), CAC, LTV / ARPU, Contribution Margin, Scenario Analysis, Target Setting, Budget Allocation, Forecasting